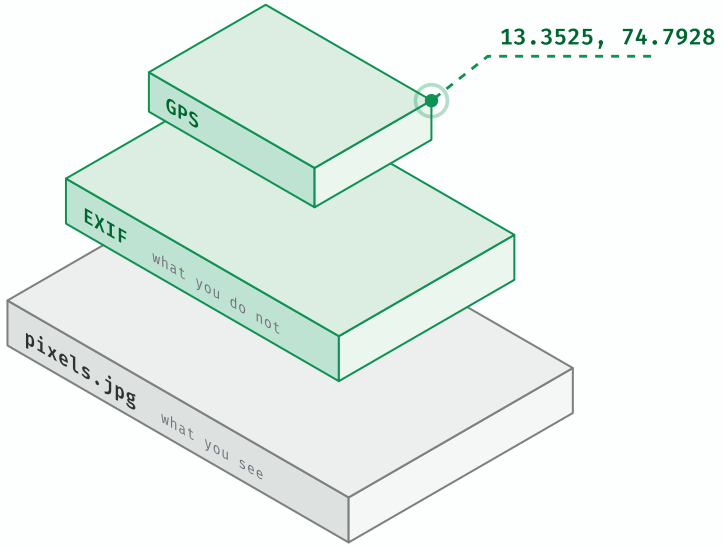
SHELL SCRIPTING / M.E. CYBERSECURITY

Your photos talk.

exifleak is a small Bash script. It reads the metadata a photo gives away, then wipes it.



ANISH SHOBITH P S / MSIS, MAHE

02 THE PROBLEM

A photo is two files in one.

Pixels you see, and an EXIF block you don't. The camera writes it. Every app that copies the file carries it along.

USUALLY KEPT

email · cloud links · AirDrop · sent as document

USUALLY STRIPPED ON UPLOAD

Instagram · Facebook · X · Signal

WHAT THE SCRIPT HUNTS

GPS Position — location
13 deg 21' 9.00" N, 74 deg 47' 34.08" E

Camera Model Name — device
iPhone 15 Pro

Serial Number — device
F2LDEMO12345

Artist — identity
Anish Shobith

Software — software
iOS 18.1

03 WHY IT MATTERS

It has already cost people.

2007 **US Army, Iraq** ↗ iTnews, 2012

Geotagged flightline photos helped attackers find four new Apache helicopters. Mortars destroyed them. The Army disclosed it in 2012.

2010 **Adam Savage** ↗ NYT via Benton, Aug 2010

He tweeted a photo of his car outside his home, captioned "off to work". The geotag gave the address.

2012 **Higinio Ochoa** ↗ Computerworld, 2012

The FBI says GPS in a photo of his girlfriend pointed to her home in Australia. He was arrested in March.

2012 **John McAfee** ↗ The Next Web, 2012

A published photo carried GPS that put him in Guatemala. He was arrested two days later, and later claimed he faked the data.

RESEARCH Friedland and Sommer, HotSec 2010: geotags on Craigslist, YouTube and Twitter revealed homes and away-times.

04 DESIGN

One job per function.

01 FILES

`for`

a file, or every image in a folder by glob

→

02 FETCH

`meta=$(exiftool)`

one call, kept in a variable

→

03 DETECT

DETECTORS

- `detect_location`
- `detect_device`
- `detect_identity`
- `detect_software`

→

04 REPORT

`show`

one `[TAG] line` each

The loop never names a check. A new check is one function and one name in `DETECTORS`.

05 THE SCRIPT

exifleak.sh 0 / 4

```
#!/bin/bash
# exifleak.sh - show what a photo gives away, or wipe it.
#   ./exifleak.sh photo.jpg folder ...      report
#   ./exifleak.sh -s photo.jpg folder ...   wipe all metadata

DETECTORS="detect_location detect_device detect_identity detect_software"

show() {   # show LABEL "Tag name"
    line=$(echo "$meta" | grep -i "^$2" | head -1 | tr -s ' ')
    if [ -n "$line" ]; then
        echo "  [$1] $line"
    fi
}

detect_location() {
    show LOCATION "GPS Position"
    maps=$(echo "$meta" | grep "^GPS Position" | cut -d: -f2- | tr -d "\"'," |
           awk '{ lat = $1 + $3/60 + $4/3600; lon = $6 + $8/60 + $9/3600
```

05 THE SCRIPT

exifleak.sh — 1 / 4 the list, and the one helper that prints

```
#!/bin/bash
# exifleak.sh - show what a photo gives away, or wipe it.
#   ./exifleak.sh photo.jpg folder ...      report
#   ./exifleak.sh -s photo.jpg folder ...   wipe all metadata

DETECTORS="detect_location detect_device detect_identity detect_software"

show() {   # show LABEL "Tag name"
    line=$(echo "$meta" | grep -i "^$2" | head -1 | tr -s ' ')
    if [ -n "$line" ]; then
        echo "  [$1] $line"
    fi
}

detect_location() {
    show LOCATION "GPS Position"
    maps=$(echo "$meta" | grep "^GPS Position" | cut -d: -f2- | tr -d "\"'," |
           awk '{ lat = $1 + $3/60 + $4/3600; lon = $6 + $8/60 + $9/3600
```

05 THE SCRIPT

exifleak.sh — 2 / 4 awk: degrees + minutes/60 + seconds/3600

```
    fi
}

detect_location() {
    show LOCATION "GPS Position"
    maps=$(echo "$meta" | grep "^GPS Position" | cut -d: -f2- | tr -d "\"'," |
           awk '{ lat = $1 + $3/60 + $4/3600; lon = $6 + $8/60 + $9/3600
                  if ($5 == "S") lat = -lat
                  if ($10 == "W") lon = -lon
                  printf "%.4f,%.4f\n", lat, lon }')
    if [ -n "$maps" ]; then
        echo "  [LOCATION] https://maps.google.com/?q=$maps"
    fi
}

detect_device() {
    show DEVICE "Camera Model"
    show DEVICE "Serial Number"
```

05 THE SCRIPT

exifleak.sh — 3 / 4 one tiny function per leak type

```
25 }
26
27 detect_device() {
28     show DEVICE "Camera Model"
29     show DEVICE "Serial Number"
30 }
31
32 detect_identity() {
33     show IDENTITY "Artist"
34     show IDENTITY "Owner Name"
35 }
36
37 detect_software() {
38     show SOFTWARE "Software"
39 }
40
41 inspect() {
42     meta=$(exiftool "$1" 2> /dev/null)
```

05 THE SCRIPT

exifleak.sh — 4 / 4 run every registered detector

```
    show SOFTWARE "Software"
}

inspect() {
    meta=$(exiftool "$1" 2> /dev/null)
    echo
    echo "$1"
    report=$(for d in $DETECTORS; do $d; done)
    if [ -z "$report" ]; then
        echo "  clean"
    else
        echo "$report"
    fi
}

scrub() {
    exiftool -overwrite_original -all= "$1" > /dev/null 2>&1
    echo "scrubbed $1"
```

06 HOW IT WORKS

One field, start to finish.

EVERY FIELD

```
exiftool "$1"
```

prints every tag as text, kept in `$meta`

```
grep -i "^Artist" | head -1
```

keeps the first line with that label

```
tr -s ' '
```

squeezes the padding

```
echo "[IDENTITY] $line"
```

prints it with its leak type

LOCATION

```
13 deg 21' 9.00" N
```

what exiftool prints

```
13 deg 21 9.00 N
```

`tr -d` drops `"` `'` `,`

```
13 + 21/60 + 9/3600
```

`awk` adds degrees, minutes, seconds; S and W go negative

```
?q=13.3525,74.7928
```

the map link

RUN

```
-s
```

picks `scrub`, otherwise `inspect`

```
for
```

hands each file, or each image in a folder, to `$action`

```
exiftool -all=
```

scrub: removes every tag in place

```
clean
```

inspect: printed when no detector reports

07 DEMO

~/photos

```
$ ./exifleak.sh demo

demo/IMG_4821.jpg
[LOCATION] GPS Position : 13 deg 21' 9.00" N, 74 deg 47' 34.08" E
[LOCATION] https://maps.google.com/?q=13.3525,74.7928
[DEVICE] Camera Model Name : iPhone 15 Pro
[DEVICE] Serial Number : F2LDEMO12345
[IDENTITY] Artist : Anish Shobith
[SOFTWARE] Software : iOS 18.1

demo/IMG_4822.jpg
[DEVICE] Camera Model Name : Pixel 8

demo/IMG_4823.jpg
clean
```

07 DEMO

~/photos

```
$ ./exifleak.sh demo

demo/IMG_4821.jpg
[LOCATION] GPS Position : 13 deg 21' 9.00" N, 74 deg 47' 34.08" E
[LOCATION] https://maps.google.com/?q=13.3525,74.7928
[DEVICE] Camera Model Name : iPhone 15 Pro
[DEVICE] Serial Number : F2LDEMO12345
[IDENTITY] Artist : Anish Shobith
[SOFTWARE] Software : iOS 18.1

demo/IMG_4822.jpg
[DEVICE] Camera Model Name : Pixel 8

demo/IMG_4823.jpg
clean
```

wipe it, then look again

```
$ ./exifleak.sh -s \
 demo/IMG_4821.jpg
scrubbed demo/IMG_4821.jpg
$ ./exifleak.sh \
 demo/IMG_4821.jpg

demo/IMG_4821.jpg
clean
```

08 LIMITS

Where it breaks.

BRITTLE BY DESIGN

It reads ExifTool's labels

A camera that writes `Model` instead of `Camera Model Name` slips past `grep`. Add a pattern, or read `exiftool -s` names.

SCOPE

One folder level

No `**`: globstar needs bash 4, and macOS ships 3.2. The script runs on both.

EXIFTOOL CAVEAT

The wipe is not total

PDF edits are incremental, so old metadata stays recoverable. `-all=` also drops orientation and colour profile on JPEGs.

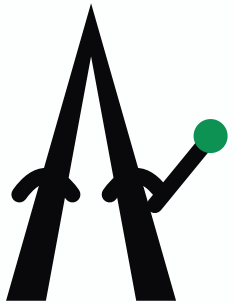
↗ ExifTool docs

PARSER RISK

The scanner is attack surface

ExifTool had RCE via DjVu (CVE-2021-22204, before 12.24) and filename injection (CVE-2022-23935, before 12.38). Stay patched.

↗ NVD



Detect → Report → Scrub

Share the scrubbed copy, never the original. Questions?

ANISHSHOBITHPS.COM / GITHUB.COM/ANISHSHOBITHPS